

PATRICK CHILUFYA BWALYA

Software Developer | AI & Web Development Specialist

Lusaka, Zambia

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Professional Summary

Highly motivated and analytical Software Developer with strong experience in full-stack development, artificial intelligence integration, and cloud technologies. Skilled in building scalable applications using modern frameworks such as React, Next.js, Laravel, and Node.js. Strong background in algorithm design, UI/UX implementation, API development, and technical troubleshooting. Possesses hands-on experience developing AI classification tools, geospatial mapping systems, and production web applications. Demonstrates excellent problem-solving skills, adaptability, and the ability to work effectively in team-based agile environments.

Technical Skills

- **Languages:** JavaScript, TypeScript, Python, PHP, Java, SQL
- **Frontend:** React, Next.js, Tailwind CSS, HTML5, CSS3, Blade, UI/UX Prototyping
- **Backend:** Laravel, Node.js, REST APIs, Authentication, Server Logic
- **Databases:** MySQL, Firebase
- **DevOps & Tools:** Git/GitHub, Docker, VS Code, Figma, Adobe Photoshop, cPanel
- **AI/ML:** TensorFlow basics, image dataset creation, model training pipelines
- **Other:** Computer repair, graphic design for digital media, shared hosting deployment

Soft Skills

- Analytical Thinking
- Communication & Collaboration
- Leadership & Initiative
- Technical Documentation
- Time Management & Task Prioritization

Certifications

- Huawei ICT Competition – Artificial Intelligence
- Huawei ICT Competition – Cloud Computing (HCIA-Cloud Computing V5.5)

Professional Experience

First World Transit Africa — Web Developer

2026 – Present

- Built a full-stack vehicle rental management platform using Laravel 12, Blade, and a custom CSS variable theme system with dark and light mode support.
- Integrated Airtel Money mobile payments including USSD push initiation, callback verification, reference and amount validation, and manual status checks.
- Implemented role-based access control (superadmin, admin, client) with middleware, Laravel Policies, and a Transaction Policy for secure rental authorization.
- Developed a queued database notification system covering rental approvals, cancellations, extensions, and new client registrations.
- Optimized performance with WebP image compression, aggregated SQL queries, database indexes, and a scheduled artisan command for automated rental expiry.

- Deployed to cPanel shared hosting — configured MySQL, storage routing without symlinks, HTTPS enforcement, and SSL. Live at firstworldtransitafrica.com.

Africa Mineral Connect — *Web Developer*

Feb 2026 – Present

- Contracted under a formal Software Development Agreement to design and build a full-stack web platform for a mineral investment business operating in the Southern African region.
- Developed a multi-role application with role-based authentication, content management, workflow automation, and a real-time messaging system.
- Built using Laravel (PHP), MySQL, and custom CSS with full mobile responsiveness.
- Deployed and configured the live production environment on cPanel shared hosting.

Academic & Personal Projects — *Software Developer*

2022 – Present

- Developed multiple full-stack and AI-powered applications including geospatial forest monitoring tools, image recognition systems, and responsive portfolio websites.
- Implemented REST APIs, handled state management, optimized user interfaces for performance, and integrated AI models into web applications.
- Regularly used Git, CI/CD basics, and agile-style development workflows.

Transparency International Zambia — *Data Collector*

2025

- Contributed to updating document tags and metadata on TI-Zambia systems, ensuring data accuracy and consistency across digital platforms.
- Helped design visual assets for internal documentation, demonstrating versatility between technical and creative tasks.

Kenneth Kaunda International Airport — *IT Intern*

2024 – 2025

- Assisted with daily management of IT systems including computer setup, troubleshooting, software updates, and maintenance of office networks.
- Gained hands-on experience diagnosing hardware issues and streamlining technical workflows within a high-security environment.

Mulungushi University — *IT Intern*

2023 – 2024

- Performed repairs on computers, printers, and networking equipment; designed banners and university media material.
- Supported staff with technical issues, improving operational workflow and ensuring minimal downtime for campus systems.

Education

Bachelor of Science in Computer Science — *Mulungushi University, Zambia*

Software Projects

- First World Transit Africa — Full-stack Laravel vehicle rental platform with Airtel Money payments, role-based access, queued notifications, and live cPanel deployment. Live at firstworldtransitafrica.com.
- Africa Mineral Connect — Multi-role Laravel platform for a mineral investment business, featuring role-based authentication, content management, workflow automation, and real-time messaging.
- PUM Portfolio Website — Client-facing consultation platform for mine agents, enabling clients to browse minerals and request consultation sessions.

- AI Tree Detection System — Web-based AI tool identifying tree species from satellite imagery using a custom-trained ML model, integrated with Leaflet maps.
- Global Forest Map Dashboard — Interactive React + Leaflet dashboard displaying tree locations, forest health statuses, and real-time conservation indicators.
- Portfolio Website — Fully responsive personal portfolio showcasing projects and skills. Live at rico-hub-code.github.io/code-hub/

References

Available upon request.